JUBAIR IBN OMAR HAMZA

M.Sc. Renewable Energy Engineering (In Progress, Exeter) | PG Diploma Data Science | B.Sc. EEE
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PERSONAL STATEMENT

Engineering researcher and graduate student with a background spanning biomedical signal processing, semiconductor device modelling, and power systems analysis, currently completing an MSc in Renewable Energy Engineering at the University of Exeter. Experienced in simulation-driven research (COMSOL, MATLAB, Python) and university teaching, with peer-reviewed publications and conference presentations to date. Seeking a research or engineering role in renewable energy or applied data science where strong analytical and modelling skills can support real-world sustainability projects.

EDUCATION

M.Sc. in Renewable Energy Engineering (In Progress) | *University of Exeter* 2027
Renewable Energy Group, College of Engineering, Mathematics and Physical Sciences.

M.Sc. in Electrical and Electronic Engineering (Incomplete/Paused) | *North South University* 2026
CGPA: 3.81/4.00
Completed Courseworks: Mobile & Wireless Communications, Robotics, Modern & Optimal Control Systems, Nanotechnology, Smart Energy Systems, Advanced VLSI Design, Solid State Electronics.

Post Graduate Diploma in Data Science | *United International University* 2024
CGPA: 3.61/4.00

B.Sc. in Electrical and Electronic Engineering | *North South University* 2023
CGPA: 3.80/4.00 (equivalent: First Class Honours) | Summa Cum Laude (Highest Distinction / First Class)

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant | *ECE Department, North South University* 09/2025 – 09/2026
- Deliver laboratory lectures, prepare lab manuals, and evaluate student performance.
- Teach four laboratory courses each semester and support BAETE-related departmental documentation.
- Courses: EEE111L, EEE141L. Taught ~140 undergrad students/semester; 420 students across 3 semesters.

Adjunct Lecturer | *EEE Department, Uttara University* 01/2026 – 05/2026
- Prepared and delivered lecture content, assignments, and examinations for 5 sections (~225 students) in one semester.
- Set, conducted, proctored, and graded quizzes, midterms, and final examinations.
- Provided timely and constructive feedback on student performance.
- Courses: EEE4137 (Communication Theory II), EEE3227 (Microprocessor & Interfacing).

Lab Instructor | *ECE Department, North South University* 07/2024 – 09/2025
- Conducted undergraduate laboratory sessions, prepared manuals, supervised experiments, and evaluated student performance.
- Courses: EEE111L, EEE241L, EEE211, EEE221, CSE231, EEE363L. Taught ~105 students/semester; 315 students across 3 semesters.

Graduate Assistant | *Math & Physics Department, North South University* 01/2024 – 07/2024
- Provided one-on-one academic guidance and evaluated assignments, quizzes, and course submissions.

Intern | *OCMS – Textile & Apparel ERP Solution* 04/2023 – 10/2023
- Contributed to ERP-related software development and automation tasks.
- Performed data analysis and programming using SQL and Python.

Undergraduate Assistant | *Math & Physics Department, North South University* 01/2023 – 12/2023
- Assisted faculty with undergraduate teaching support, grading, and student guidance in mathematics and physics.

Front-of-House Helper | *Crispyon Restaurant, Uttara, Dhaka* Casual
- Served customers, took orders, and supported kitchen and stock tasks at a family-run restaurant.
- Maintained a friendly, customer-facing presence during peak service periods and adapted to shifting demands as part of a small team.

RESEARCH EXPERIENCE

Research Assistant | *Dr. Zaid Bin Mahbub, North South University* 05/2024 – 09/2026
- Conducting research on Pulsed Electromagnetic Field (PEMF) devices and their impact on peripheral circulation.
- Simulation and modelling using Python (NumPy, SciPy, PyVista, VTK, Plotly), MATLAB, and COMSOL Multiphysics.
- Assisted in developing a prototype biomedical monitoring system using Raspberry Pi 5 with multiple sensors, involving sensor interfacing, data acquisition, and real-time signal processing.

Research Volunteer | *Cyber-Physical Systems (CPS) Lab, North South University* 05/2025 – 02/2026
- Contributed to power system modelling and analysis using PSAT, including design and validation of a 94-bus test system.

- Performed dynamic fault and stability analysis using TSAT to study transient system behavior under disturbance scenarios.
- Gained experience in large-scale system modelling, power system dynamics, and simulation-driven analysis.

RESEARCH & PUBLICATIONS

[1] *"Prediction of Conduction Velocity Distribution and Motor Unit Action Potential from Evoked Compound Muscle Action Potential: A Novel Metaheuristic-Based Framework."* **Biomedical Physics & Engineering Express**, Vol. 12, Article 045044, IOP Publishing, 2026. DOI: 10.1088/2057-1976/ae8af8.

[2] *"Pulsed electromagnetic field driven perturbation of peripheral arterial hemodynamics: integrated evidence from Doppler ultrasound and photoplethysmographic signal analysis."* **Biomedical Physics & Engineering Express**, Vol. 12, Article 055065, IOP Publishing, 2026. DOI: 10.1088/2057-1976/aeb382.

[3] *"Geometry-Dependent Hemodynamic Responses to Pulsed Electromagnetic Field Exposure in the Radial Artery."* **Bioelectromagnetics** (Wiley), Under Review, 2026.

CONFERENCE PRESENTATIONS

International Conference on Physics in Medicine (ICPM 2025) 2025
Modelling and Simulation of Pulsed Electromagnetic Field (PEMF) Effects on Peripheral Arterial Blood Flow via Photoplethysmography (PPG)

International Conference on Advances in Physics (ICAP) 2025
Simulation-Based Investigation of Pulsed Electromagnetic Field (PEMF) Effects on Hemodynamics via Physics-Informed Modeling

2nd Intl. Conference on Frontiers in Science: Innovation & Technology for Greener Industry (ICFS: ITGI 2025) 2025
Magnetohydrodynamic Modulation of Peripheral Blood Flow under Pulsed Electromagnetic Field: An In-Vitro Model Study

Bangladesh Physical Society International Conference on Physics (2024) 2024
Impacts of PEMF Applications on Photoplethysmography Responses

Symposium on Simulation for Everything: The Future of Digital Innovation 2026
Poster presentation, North South University (AUD 801 & NSU Plaza), 12 June 2026

CONFERENCES & ACADEMIC EVENTS ATTENDED

Biophysical Society Networking Event 2025
Bridging the Theoretical and Experimental Biophysicists in Bangladesh

TECHNICAL SKILLS

Simulation & Modelling	COMSOL Multiphysics, MATLAB, LTSpice, Multisim, Cadence, PSAT, TSAT
Programming & Data Science	Python (NumPy, SciPy, PyVista, VTK, Plotly), SQL, Jupyter, PyCharm, GitHub, Google Colab
Hardware & Systems	Arduino, Raspberry Pi, PCB/VLSI Design, Embedded Systems, Digital Logic Design
Research & Analysis	Biomedical Signal Processing, Electromagnetics, Nanotechnology, Microelectronics

SPECIALTY AREAS

- Power Systems & Grid Analysis (PSAT/TSAT, fault & stability studies)
- Control Systems (Modern & Optimal Control, Robotics)
- Biomedical Signal Processing (Doppler, PPG, CMAP, PEMF)
- Embedded & IoT Systems (Raspberry Pi, sensor interfacing, real-time acquisition)

AWARDS & SCHOLARSHIPS

- Talent Pool Scholarship, JSC – Cambrian School & College.
- Summa Cum Laude (Highest Distinction) – B.Sc. in EEE, North South University.
- 100% Merit-Based Scholarship and GTA-ship – M.Sc. in EEE, North South University.

SHORT COURSES & CERTIFICATIONS

- Data Science Short Course – Dr. Jennifer Widom, Professor & Dean of Engineering, Stanford University.
- MS Office Comprehensive Online Course – BASIS Institute of Technology & Management Limited and US Software Limited.

LANGUAGES

Bengali: Native **English:** Fluent — IELTS Academic, Overall Band 8.0 (Listening 8.0, Reading 8.5, Writing 7.0, Speaking 8.5), CEFR C1.